# Master Formula Inventory

This file contains all formulas physically extracted from source documents.

## From Source: `formulas.typ`

**Variable: `flat_sums_btherom`**

Display:

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}...\left[\sum_{n_l=0}^{n_{l-1}}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{l \text{ sums}}$$

**Variable: `falling_sums_btherom`**

Display: for all $a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid d$ in $\mathbb{R}$ where $\left(\frac{x}{d} = l, l \in \mathbb{Z}^+\right) \Rightarrow$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}...\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]...a^{b-c}\right]\right]a^{r-b}\right]}^{l \text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}...\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}[(x-la)^{n_{l-1}}]\right]...a^{b-c}\right]\right]a^{r-b}\right]}^{l-1 \text{ sums}}$$

$$+...+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}[(x-2a)^b]a^{r-b}\right]}^{1 \text{ sums}}$$

$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

**Variable: `falling_sum_lr_btherom`**

Display:

$$\text{for all } a \in \mathbb{R} \mid r \in \mathbb{Z}^+ \mid l \in \mathbb{Z}^+$$

$$x^r = \sum_{b=0}^{l-1}\left[\sum_{c=0}^{r-1}\left[\binom{r}{c}(x-(b+1)a)^c a^{r-c}\right]\right] + (x-la)^r$$

**Variable: `n_falling_sums_freedom_btherom`**

Display:

$$\text{for all } r \in \mathbb{Z}^+ \mid n_0 \in R \mid n_0 \neq 0 \mid n_1...n_n \in \mathbb{Z}^+$$

$$x^r = -1\sum_{b_n=0}^{n_n-1}\left[...\sum_{b_2=0}^{n_2-1}\left[\sum_{b_1=0}^{n_1-1}\left[\sum_{b_0=0}^{r-1}\left[\binom{r}{b_0}(((x+b_n n_n...n_1n_0)+...+b_2n_1n_0)+b_1n_0)^{b_0}n_0^{r-b_0}\right]\right]\right]\right] + ((x+n_n...n_2n_1n_0)^r$$